

hexatic graphene

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1 Correlation function

The Hamiltonian of the system is given by [1]:

$$H = \frac{\epsilon}{\rho} \sum_i (\theta_i^2) + \frac{\gamma}{3\sqrt{\rho}} \sum_i \sum_{j \in N_i} (|\theta_i - \theta_j|)(A - \ln(|\theta_i - \theta_j|))$$

where ϵ is a material-dependent constant, ρ is the density of grains, γ and A are some constants. N_i is the set of neighbors of grain i .

The partition function is given by:

$$\mathcal{Z} = \int \prod_i d\theta_i \exp \left(-\beta \sum_i E_i \right) = \int \prod_i d\theta_i \text{Tr} \{ \exp(-\beta H) \}$$

where $\beta = \frac{1}{k_B T}$, k_B is the Boltzmann constant and T is the temperature.

1.1 Large grain limit

In the limit of large grains, global alignment with substrate dominates grains boundaries energy, and the mis-orientation distribution becomes distance-independent

$$\mathcal{Z} = \int \prod_i d\theta_i \exp \left(-\beta \frac{\epsilon}{\rho} \sum_i \theta_i^2 \right) = \prod_i \int d\theta_i \exp \left(-\beta \frac{\epsilon}{\rho} \theta_i^2 \right)$$

So we have a gaussian distribution for each grain orientation θ_i with mean 0 and variance $\sigma^2 = \frac{\rho}{2\beta\epsilon}$. The correlation function is given by:

$$C(\theta_i, \theta_j) = \langle \phi_i \phi_j^* \rangle = \langle e^{i6(\theta_i - \theta_j)} \rangle$$

As the angles are independent, we have:

$$C(\theta_i, \theta_j) = |\langle e^{i6\theta_i} \rangle|^2 = |e^{-18\sigma^2}|^2 = e^{-18\frac{\rho}{\beta\epsilon}}$$

1.2 Small grain limit

In the limit of small grains, grain boundaries energy dominates global alignment with substrate. To compute the correlation function, the Kosterlitz-Thouless and Wegner [2] approaches are used.

The goal is to approximate the hamiltonian to a quadratic form. However, the function

$$|\theta_i - \theta_j|(A - \ln(|\theta_i - \theta_j|))$$

is not differentiable at 0. To overcome this issue, we approximate it by a differentiable function $f(x)$ such that $f(x) \approx x(A - \ln(x))$ for $x \gg 0$ and $f(x) \approx 0$ for $x \approx 0$. A possible choice is

$$f(x) = \sqrt{x^2 + a^2} \left(A - \ln \left(\sqrt{x^2 + a^2} \right) \right)$$

where a is a small parameter.

We have:

$$H = \frac{\epsilon}{\rho} \sum_i \theta_i^2 + \frac{\gamma}{3\sqrt{\rho}} \sum_i \sum_{j \in N_i} \left(\sqrt{|\theta_i - \theta_j|^2 + a^2} \right) \left(A - \frac{1}{2} \ln (|\theta_i - \theta_j|^2 + a^2) \right)$$

Let $g_a(\Delta\theta)$ be the function defined by:

$$g_a(\Delta\theta) = \left(\sqrt{|\Delta\theta|^2 + a^2} \right) \left(A - \frac{1}{2} \ln (|\Delta\theta|^2 + a^2) \right)$$

For a small $\Delta\theta$ and assume $\Delta\theta \geq 0$, we can expand g_a :

$$g_a(\Delta\theta) = cste + \frac{\Delta\theta^2}{2a} (A - \ln a - 1) + \mathcal{O}(\Delta\theta^4)$$

So the grain boundary term can be approximated by a quadratic term for small misorientation angles. The hamiltonian becomes:

$$H = \int d^2r \left[\frac{\mu}{2} \theta^2 + \frac{K}{2} (\nabla\theta)^2 \right]$$

where $\mu = \frac{2\epsilon}{\ell^2\rho}$, $K = \frac{\gamma}{3\sqrt{\rho}} \frac{z}{2a} (A - \ln a - 1)$, ℓ is the average grain size and z is the coordination number of the lattice.

This is exactly an Hamiltonian of a massive gaussian field. In this case, we have:

$$\langle (\theta(\mathbf{r}) - \theta(0))^2 \rangle = \frac{k_B T}{2\pi K} \left(\ln \frac{r}{\xi} + K_0 \frac{r}{\xi} \right)$$

where $\xi = \sqrt{\frac{K}{\mu}}$ is the correlation length.

The correlation function is given by:

$$G_6(r) = \langle e^{i6(\theta(\mathbf{r}) - \theta(0))} \rangle = e^{-18 \langle (\theta(\mathbf{r}) - \theta(0))^2 \rangle}$$

So we have:

$$G_6(r) = \begin{cases} \left(\frac{r}{\ell} \right)^{-\eta_6} & \ell \ll r \ll \xi \\ \left(\frac{\xi}{\ell} \right)^{-\eta_6} e^{-\eta_6 K_0 r / \xi} & r \gg \xi \end{cases}$$

where $\eta_6 = \frac{18k_B T}{2\pi K}$.

References

- [1] Oleg V. Yazyev and Steven G. Louie. “Topological defects in graphene: Dislocations and grain boundaries”. In: *Physical Review B* 81 (19 May 2010), p. 195420. ISSN: 1098-0121. DOI: 10.1103/PhysRevB.81.195420. URL: <https://link.aps.org/doi/10.1103/PhysRevB.81.195420>.
- [2] Franz Wegner. “Spin-ordering in a planar classical Heisenberg model”. In: *Zeitschrift fur Physik* 206 (5 Oct. 1967), pp. 465–470. ISSN: 1434-6001. DOI: 10.1007/BF01325702. URL: <http://link.springer.com/10.1007/BF01325702>.